

2.5 STOCHASTIC PROCESSES

A stochastic process, also called a random process, is a very simple generalization of the concept of an RV. A stochastic process $X(t)$ is an RV X that changes with time.⁴ A stochastic process can be one of four types.

- If the RV at each time is continuous and time is continuous, then $X(t)$ is a continuous random process. For example, the temperature at each moment of the day is a continuous random process because both temperature and time are continuous.
- If the RV at each time is discrete and time is continuous, then $X(t)$ is a discrete random process. For example, the number of people in a given building at each moment of the day is a discrete random process because the number of people is a discrete variable and time is continuous.
- If the RV at each time is continuous and time is discrete, then $X(t)$ is a continuous random sequence. For example, the high temperature each day is a continuous random sequence because temperature is continuous but time is discrete (day one, day two, etc.).
- If the RV at each time is discrete and time is discrete, then $X(t)$ is a discrete random sequence. For example, the highest number of people in a given building each day is a discrete random sequence because the number of people is a discrete variable and time is also discrete.

Since a stochastic process is an RV that changes with time, it has a distribution and density function that are functions of time. The PDF of $X(t)$ is

$$F_X(x, t) = P(X(t) \leq x) \quad (2.77)$$

If $X(t)$ is a random vector, then the inequality above is an element-by-element inequality. For example, if $X(t)$ has n elements, then

$$F_X(x, t) = P[X_1(t) \leq x_1 \text{ and } \cdots X_n(t) \leq x_n(t)] \quad (2.78)$$

The pdf of $X(t)$ is

$$f_X(x, t) = \frac{dF_X(x, t)}{dx} \quad (2.79)$$

If $X(t)$ is a random vector, then the derivative above is taken once with respect to each element of x . For example, if $X(t)$ has n elements, then

$$f_X(x, t) = \frac{d^n F_X(x, t)}{dx_1 \cdots dx_n} \quad (2.80)$$

The mean and covariance of $X(t)$ are also functions of time:

$$\bar{x}(t) = \int_{-\infty}^{\infty} x f(x, t) dx$$

⁴Actually, the independent variable does not have to be time; for example, it could be spatial location or something else. But typically the independent variable is time, and in this book it will always be time.

$$\begin{aligned}
 C_X(t) &= E \left\{ [X(t) - \bar{x}(t)] [X(t) - \bar{x}(t)]^T \right\} \\
 &= \int_{-\infty}^{\infty} [x - \bar{x}(t)] [x - \bar{x}(t)]^T f(x, t) dx
 \end{aligned} \tag{2.81}$$

Note that $X(t)$ at two different times (t_1 and t_2) comprise two different random variables $[X(t_1)$ and $X(t_2)]$. Therefore, we can talk about the joint distribution and joint density functions of $X(t_1)$ and $X(t_2)$. These are called the second-order distribution function and the second-order density function:

$$\begin{aligned}
 F(x_1, x_2, t_1, t_2) &= P(X(t_1) \leq x_1, X(t_2) \leq x_2) \\
 f(x_1, x_2, t_1, t_2) &= \frac{\partial^2 F(x_1, x_2, t_1, t_2)}{\partial x_1 \partial x_2}
 \end{aligned} \tag{2.82}$$

As discussed earlier, if $X(t)$ is an n -element random vector, then the inequality that defines $F(x_1, x_2, t_1, t_2)$ actually consists of $2n$ inequalities, and the derivative that defines $f(x_1, x_2, t_1, t_2)$ actually consists of $2n$ derivatives.

The correlation between the two RVs $X(t_1)$ and $X(t_2)$ is called the autocorrelation of the stochastic process $X(t)$:

$$R_X(t_1, t_2) = E [X(t_1)X^T(t_2)] \tag{2.83}$$

The autocovariance of a stochastic process is defined as

$$C_X(t_1, t_2) = E \left\{ [X(t_1) - \bar{X}(t_1)] [X(t_2) - \bar{X}(t_2)]^T \right\} \tag{2.84}$$

For some stochastic processes, the pdf does not change with time. For example, if we flip a coin ten times then we can view that process as a stochastic process with the statistics of the process being the same at each of the ten time instances. In this case, the stochastic process is called strict-sense stationary (SSS), or just stationary for short. In this case, the mean of the stochastic process is constant with respect to time, and the autocorrelation is a function of the time difference $t_2 - t_1$ (not a function of the absolute times):

$$\begin{aligned}
 E[X(t)] &= \bar{x} \\
 E[X(t_1)X^T(t_2)] &= R_X(t_2 - t_1)
 \end{aligned} \tag{2.85}$$

For some stochastic processes, these two conditions are true even though the pdf does change with time. Stochastic processes for which these two conditions are true are called wide-sense stationary (WSS). A stationary process is wide-sense stationary, but a wide-sense stationary process may or may not be stationary. From the definition of autocorrelation, it can be shown that for a wide-sense stationary process the following properties hold:

$$\begin{aligned}
 R_X(0) &= E[X(t)X^T(t)] \\
 R_X(-\tau) &= R_X(\tau)
 \end{aligned} \tag{2.86}$$

For scalar stochastic processes, it can be shown that

$$|R_X(\tau)| \leq R_X(0) \tag{2.87}$$

■ EXAMPLE 2.12

1. The high temperature each day can be considered a stochastic process. However, this process is not stationary. The high temperature on a day in July might be an RV with a mean of 100 degrees Fahrenheit, but the high temperature on a day in December might have a mean of 30 degrees. This is a stochastic process whose statistics change with time, so the process is not stationary.
2. Electrical noise in a voltmeter might have a mean of zero and a variance of one millivolt. If we come back the next day and measure the noise again, the mean and variance may be the same as before. If the statistics of the noise are the same every day, then the electrical noise is a stationary process. Note that in reality the noise statistics will eventually change. For example, after a few decades the instrument will begin degrading and the electrical noise mean and variance will change. In this sense, there is no such thing as a stationary random process. Eventually, the universe will freeze and all signals will change. But for practical purposes, if the statistics of a random process do not change over the time interval of interest, then we consider the process to be stationary.
3. Tomorrow's closing price of the Dow Jones Industrial Average might be an RV with a certain mean and variance. However, 100 years ago the closing price had a mean that was much lower. The closing price of the stock market is an RV whose mean generally increases with time. Therefore, the stock market price is a nonstationary stochastic process.

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Suppose we have a stochastic process $X(t)$. Further suppose that the process has a realization $x(t)$. The time average of $X(t)$ is denoted as $A[X(t)]$, and the time autocorrelation of $X(t)$ is denoted as $R[X(t)]$. These quantities are defined for continuous-time random processes as

$$\begin{aligned} A[X(t)] &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt \\ R[X(t), \tau] &= A[X(t)X^T(t + \tau)] \end{aligned} \quad (2.88)$$

The definitions for discrete-time random processes are straightforward extensions of the continuous-time definitions.

An ergodic process is a stationary random process for which

$$\begin{aligned} A[X(t)] &= E(X) \\ R[X(t), \tau] &= R_X(\tau) \end{aligned} \quad (2.89)$$

In the real world, we are often limited to only a few realizations of a stochastic process. For example, if we measure the fluctuation of a voltmeter reading, we are actually only measuring only one realization of a stochastic process. We can compute the time average, time autocorrelation, and other time-based statistics of the realization. If the random process is ergodic, then we can use those time averages to estimate the statistics of the stochastic process.